Vulnerability Analysis Report

Name: Muhammad Usman Saleem

Course: Certified Ethical Hacker (CEH)

Lab: Week 4 - Vulnerability Analysis

1. Introduction

This lab focuses on performing a vulnerability assessment on a simulated network environment. The objective is to identify security weaknesses in network services, servers, and web applications, and provide recommendations to mitigate them.

2. Objectives

- Identify vulnerabilities using scanning tools

- Analyze risks based on severity levels

- Provide remediation steps

3. Tools Used

- OpenVAS

- Nmap

- Nessus Essentials

- Nikto

4. Methodology

The following steps were performed during the lab:

• Setup and run vulnerability scanners

• Scan target system (Metasploitable)

• Analyze scan results

• Identify vulnerabilities and classify risk

5. Screenshots

Screenshot 1:

```
┌──(kali㉿Usman)-[~]
└─$ nmap --script=vuln 192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-16 06:46 -0400
Nmap scan report for 192.168.1.5
Host is up (0.010s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
|_sslv2-drown: ERROR: Script execution failed (use -d to debug)
| ssl-poodle:
|   VULNERABLE:
|   SSL POODLE information leak
|     State: VULNERABLE
|     IDs:  BID:70574  CVE:CVE-2014-3566
|           The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
|           products, uses nondeterministic CBC padding, which makes it easier
|           for man-in-the-middle attackers to obtain cleartext data via a
|           padding-oracle attack, aka the "POODLE" issue.
|     Disclosure date: 2014-10-14
|     Check results:
|       TLS_RSA_WITH_AES_128_CBC_SHA
|     References:
|       https://www.imperialviolet.org/2014/10/14/poodle.html
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-3566
|       https://www.securityfocus.com/bid/70574
|_      https://www.openssl.org/~bodo/ssl-poodle.pdf
| smtp-vuln-cve2010-4344:
|_  The SMTP server is not Exim: NOT VULNERABLE
| ssl-dh-params:
|   VULNERABLE:
|   Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
|     State: VULNERABLE
|       Transport Layer Security (TLS) services that use anonymous
|       Diffie-Hellman key exchange only provide protection against passive
|       eavesdropping, and are vulnerable to active man-in-the-middle attacks
|       which could completely compromise the confidentiality and integrity
|       of any data exchanged over the resulting session.
|     Check results:
|       ANONYMOUS DH GROUP 1
|             Cipher Suite: TLS_DH_anon_EXPORT_WITH_DES40_CBC_SHA
|             Modulus Type: Safe prime
|             Modulus Source: Unknown/Custom-generated
|             Modulus Length: 512
|             Generator Length: 8
|             Public Key Length: 512
|     References:
|       https://www.ietf.org/rfc/rfc2246.txt
|
```

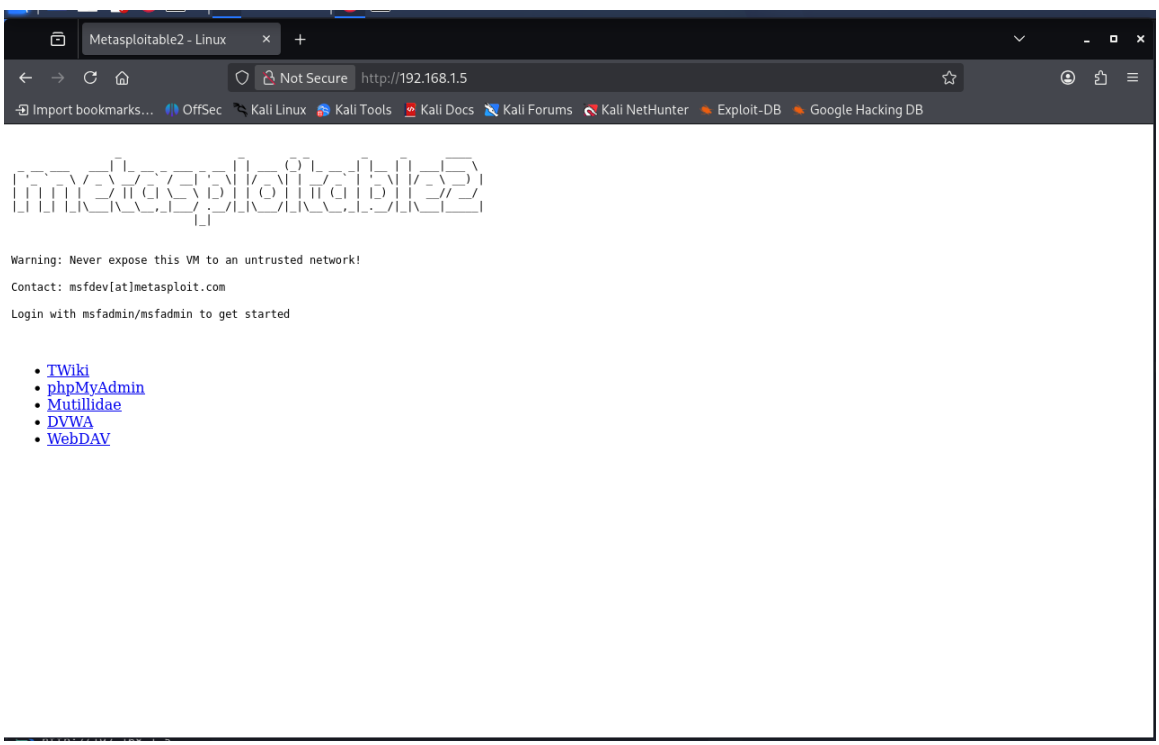
Screenshot 2:

```
┌──(kali㉿Usman)-[~]
└─$ nmap --script=http-vuln*  192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-16 07:00 -0400
Nmap scan report for 192.168.1.5
Host is up (0.047s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
|_http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:EA:40:83 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 78.49 seconds
```

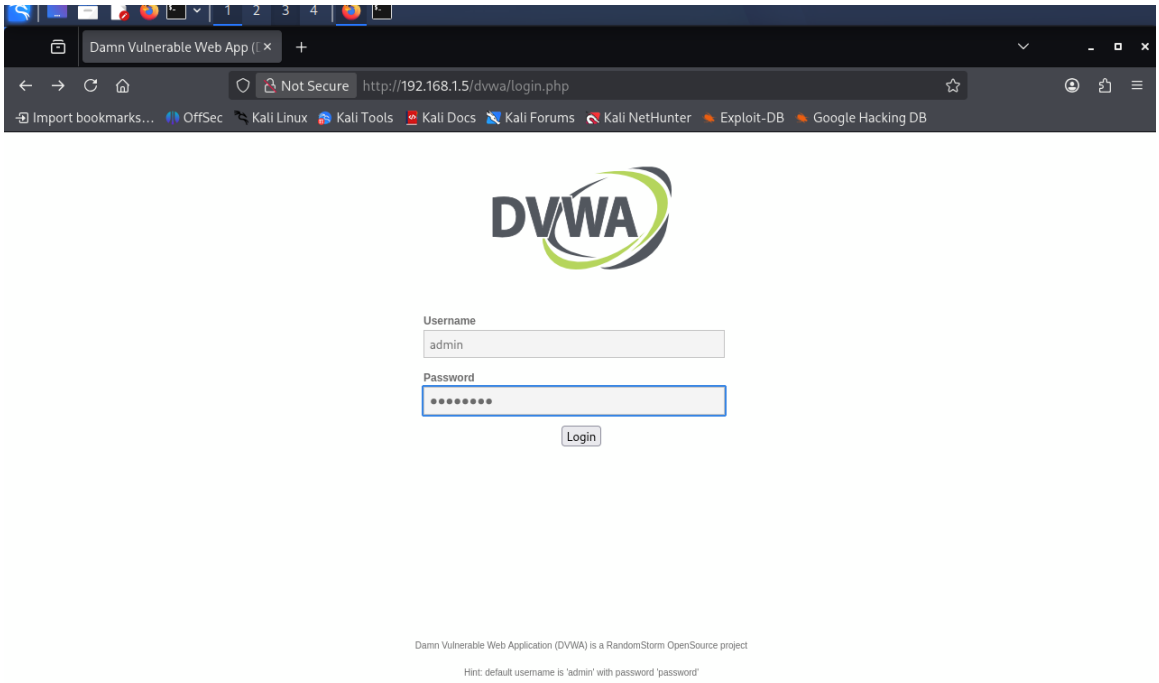
Screenshot 3:

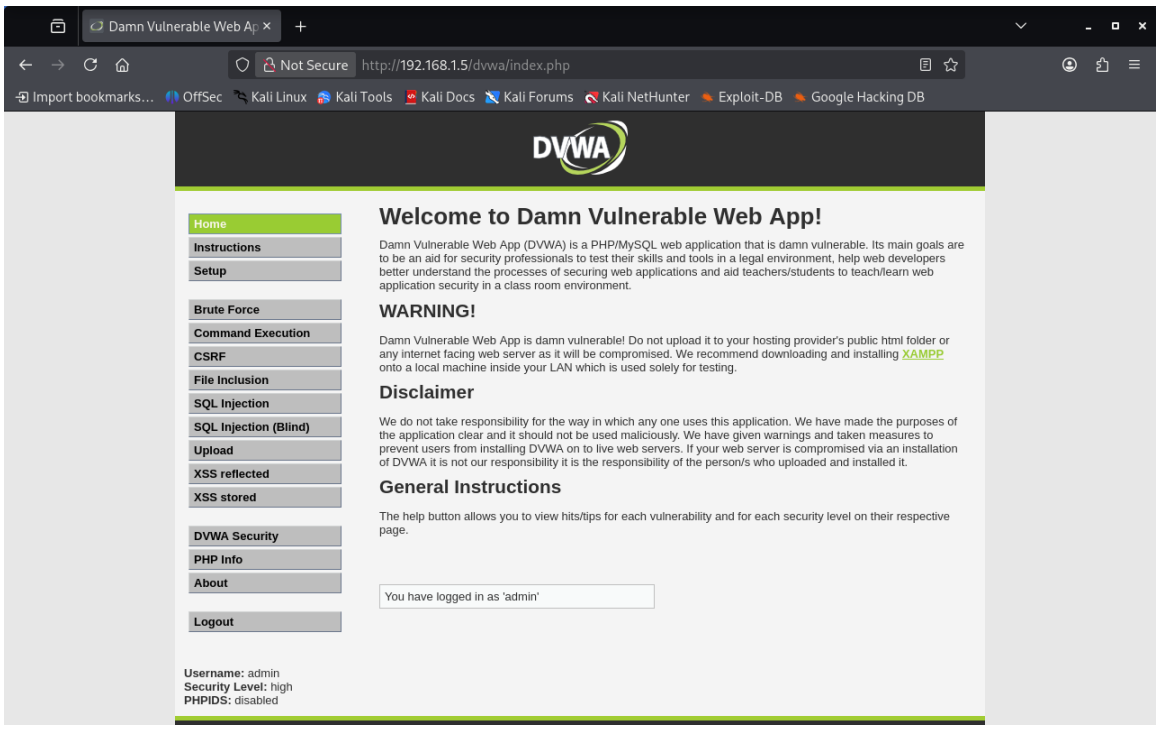
```
┌──(kali㉿Usman)-[~]
└─$ nmap --script=ftp-anon 192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-16 07:12 -0400
Nmap scan report for 192.168.1.5
Host is up (0.034s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:EA:40:83 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.87 seconds
```

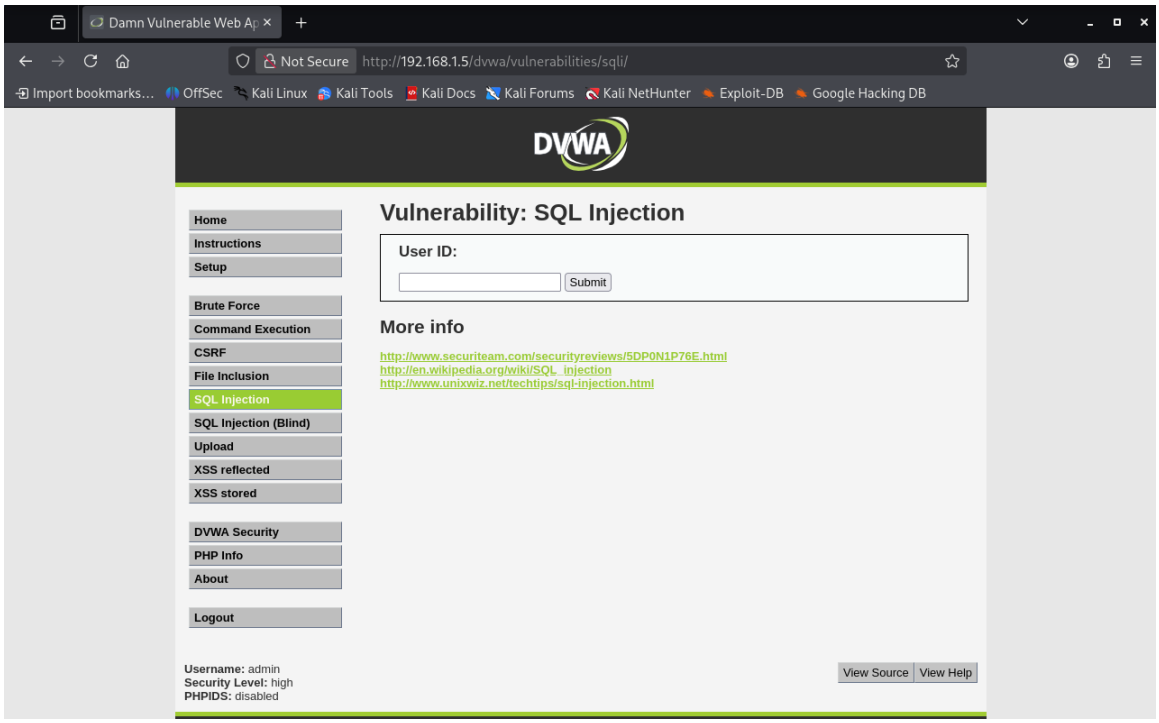
Screenshot 4:

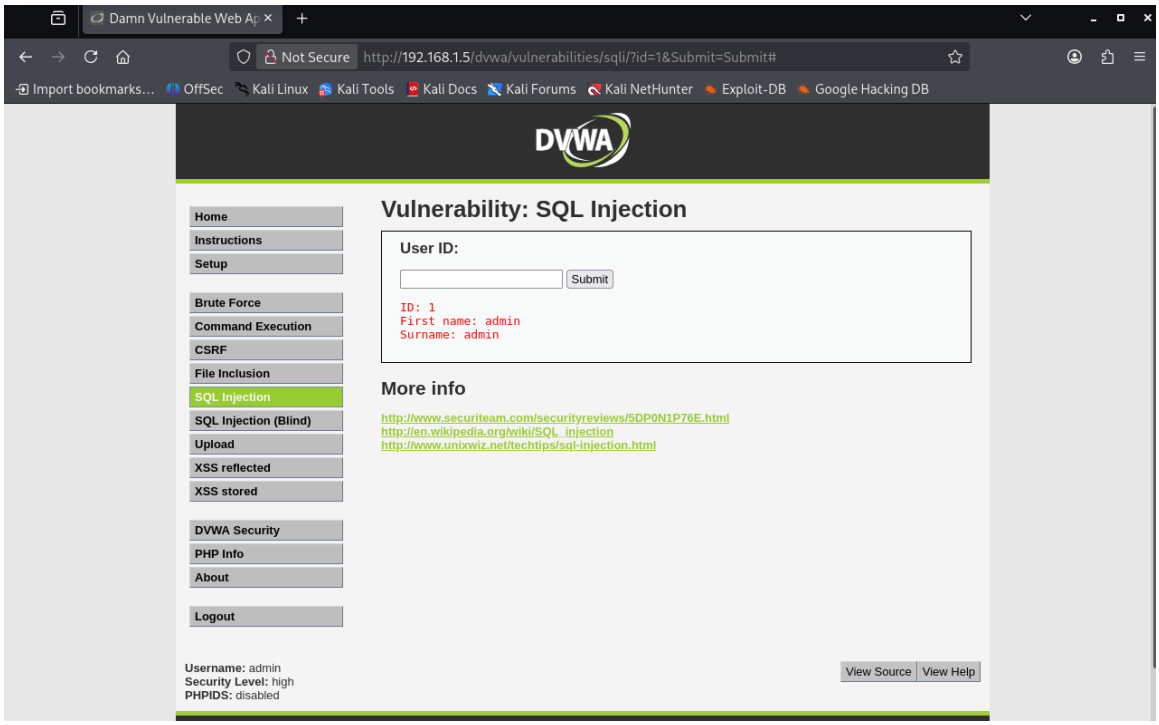


Screenshot 5:

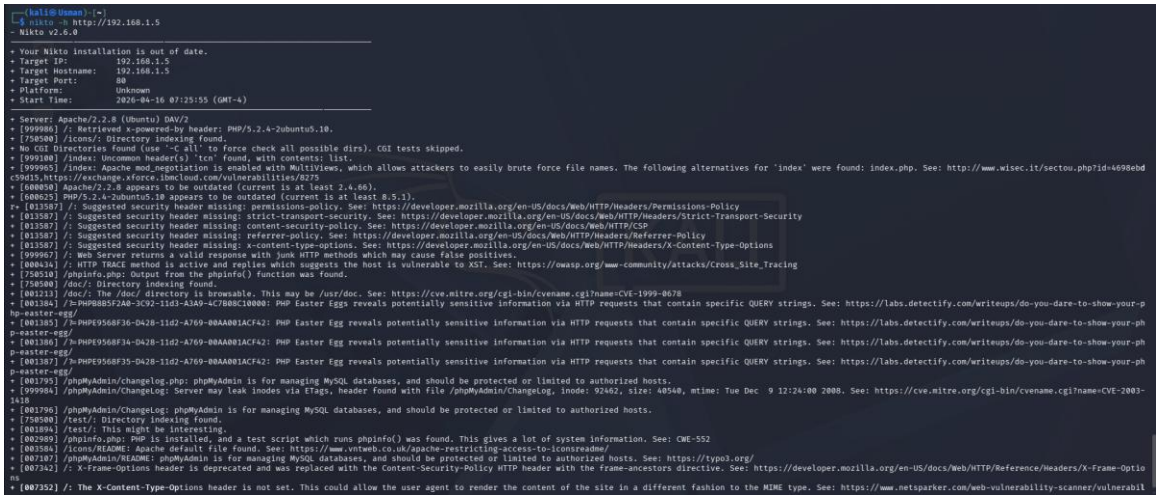








Screenshot 6:



```
┌──(kali㉿Usman)-[~]
└─$ nikto -h http://192.168.1.5
- Nikto v2.6.0
---------------------------------------------------------------------------
+ Your Nikto installation is out of date.
+ Target IP:          192.168.1.5
+ Target Hostname:    192.168.1.5
+ Target Port:        80
+ Platform:           Unknown
+ Start Time:         2026-04-16 07:25:55 (GMT-4)
---------------------------------------------------------------------------
+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ [999986] /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ [750500] /icons/: Directory indexing found.
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [999100] /index: Uncommon header(s) 'tcn' found, with contents: list.
+ [999965] /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=4698ebd
c59d15,https://exchange.xforce.ibmcloud.com/vulnerabilities/8275
+ [600050] Apache/2.2.8 appears to be outdated (current is at least 2.4.66).
+ [600625] PHP/5.2.4-2ubuntu5.10 appears to be outdated (current is at least 8.5.1).
r+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [999967] /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ [000434] /: HTTP TRACE method is active and replies which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross_Site_Tracing
+ [750510] /phpinfo.php: Output from the phpinfo() function was found.
+ [750500] /doc/: Directory indexing found.
+ [001213] /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ [001384] /?=PHPB8B5F2A0-3C92-11d3-A3A9-4C7B08C10000: PHP Easter Eggs reveals potentially sensitive information via HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/writeups/do-you-dare-to-show-your-p
hp-easter-egg/
+ [001385] /?=PHPE9568F36-D428-11d2-A769-00AA001ACF42: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/writeups/do-you-dare-to-show-your-ph
p-easter-egg/
+ [001386] /?=PHPE9568F34-D428-11d2-A769-00AA001ACF42: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/writeups/do-you-dare-to-show-your-ph
p-easter-egg/
+ [001387] /?=PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP Easter Egg reveals potentially sensitive information via HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/writeups/do-you-dare-to-show-your-ph
p-easter-egg/
+ [001795] /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ [999984] /phpMyAdmin/ChangeLog: Server may leak inodes via ETags, header found with file /phpMyAdmin/ChangeLog, inode: 92462, size: 40540, mtime: Tue Dec  9 12:24:00 2008. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-
1418
+ [001796] /phpMyAdmin/ChangeLog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ [750500] /test/: Directory indexing found.
+ [001894] /test/: This might be interesting.
+ [002989] /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ [003584] /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ [007107] /phpMyAdmin/README: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts. See: https://typo3.org/
+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Policy HTTP header with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/X-Frame-Optio
ns
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabil
```

6. Findings

List of vulnerabilities identified during the scan:

- FTP Anonymous Login Enabled

- Outdated Apache Version

- Open Telnet Port

- SMB Null Session Allowed

7. Risk Analysis Table

Vulnerability	Severity	CVSS Score	Recommended Fix
FTP Anonymous Login	Critical	9.8	Disable anonymous login
Outdated Apache	High	7.5	Upgrade Apache
Open Telnet Port	High	7.2	Disable Telnet, use SSH
SMB Null Session	Medium	5.0	Disable null sessions

8. Conclusion

The vulnerability assessment revealed multiple security issues in the target system. Critical and high-risk vulnerabilities must be addressed immediately to prevent exploitation.